IFS Projects for RDi

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Overview

The IFS Projects for RDi plug-in is used for managing RPGLE projects stored in the IFS:

- Subsystem IFS Projects in the Remote Systems view.
- Building projects and files with TOBi (The Object Builder for i).
- Building projects and files with gmake.
- SSH terminal view.
- Checking the syntax of RPGLE and SQLRPGLE files, with the errors shown in the Error List view of RDi.
- Running the build commands in the SSH terminal or in a job of the RSE connection, with the errors shown in the Error List view.

Menu Options

The menu options are in the context menus of the *Remote Systems* view:

- *Open SSH Terminal* on an IBM i connection, *Open SSH Terminal Here* on an IFS directory, file or filter.
- *New > IFS Project...* on the subsystem *IFS Projects* and on IFS directories.
- *Build* on a project or a file of a project.
- *Check Syntax* on an RPGLE or SQLRPGLE file of a project.
- *Open SSH Terminal in Project Directory* on a project.
- *Properties > IFS Project* on a project edits *iproj.json*.
- *Properties > IFS Project > Environment* edits the environment variables of the project.

Preferences

The preferences page is labeled as *IFS Projects*. It contains the settings of the SSH connection and of the terminals and the defaults of the IFS projects.

Run build and compile commands selects, where Build runs: in the SSH terminal of the project (default) or in a job of the RSE connection.

The preset sets the default build and compile command for gmake or TOBi. Each control has a tool tip with a short description.

A more detailed description can be found on the IFS Projects project page on GitHub.

SSH keys are configured on page General > Network Connections > SSH2. JSch supports RSA and ECDSA keys in PEM format, but no Ed25519 keys.

Syntax Check

Check Syntax compiles an RPGLE or SQLRPGLE stream file of an IFS project with option *EVENTF and shows the errors in the Error List view of RDi. A double-click on an error opens the source file or the copy book at the line of the error.

The file is compiled in a job of its own, which is set up like the build jobs of TOBi:

- The current directory is the project directory.

- The commands of `setIBMiEnvCmd` are executed and the libraries of `curlib`, `preUsrlibl` and `postUsrlibl` are added to the library list. Variables, such as `&ENVCMD`, are replaced by the environment variables of the project.
- The directories of `includePath` are passed as include directories.

RPGLE files are compiled with CRTBNDRPG or, for NOMAIN modules, with CRTRPGMOD and option *NOGEN, so no object is created. SQLRPGLE files are compiled with CRTSQLRPGI into a module in QTEMP, because *NOGEN would skip the RPG compiler.

If the file has unsaved changes, you are asked whether to save them before the syntax check.

LPEX Editor

In the LPEX editor, the syntax check is available as LPEX user action. The plug-in does not change the LPEX profile. Add the user action on the LPEX preference pages:

1. LPEX Editor > User Actions: name `ifsCheckSyntax`, class `de.tools400.ifsprojects.lpex.CheckSyntaxAction`.
2. LPEX Editor > User Key Actions: assign a key to action `ifsCheckSyntax`.

The action can also be run on the LPEX command line:

```
action ifsCheckSyntax
```

Build in the SSH Terminal

By default, Build runs in the SSH terminal of the project, which shows the output. Afterwards the errors of the compiled objects are shown in the Error List view, as described in Build in a Job of the RSE Connection.

To learn when the build has ended, the plug-in runs the command in a subshell. The subshell writes marker files into the directory of temporary files of RDi (`<home>/.eclipse/RSE`), also when the build is interrupted with Ctrl+C or the terminal is closed. Therefore the command line in the terminal is longer than the build command. While the plug-in waits for the end of the build, a job is shown in the Progress view. Canceling it only stops waiting; interrupt the build with Ctrl+C in the terminal.

Build in a Job of the RSE Connection

If Run build and compile commands is set to In a job of the RSE connection, Build runs in a job of the RSE connection, without SSH login. The output is shown line by line in the Commands Log view. Afterwards the errors of the compiled objects are shown in the Error List view. The SSH terminal remains available for interactive work.

The objects must be compiled with option *EVENTF. TOBi writes the events files into directory .evfevent of the project. For gmake builds, the events file EVFEVENT of the library specified as objlib (or curlib) in iproj.json is read.

Builds of the same project run one after the other. Cancel ends the job of the RSE connection; jobs started by the build may still be running on the server.

Change Log 1.0.0.r (11.10.2026)

Plug-in: 9.6.0.13+

- Initial version

IFS Projects On The Internet

Project page on GitHub	https://github.com/tools-400/ifs-terminal-plugin/
Homepage	https://tools-400.github.io/ifs-terminal/
RDi 9.6+ Update Site	https://tools-400.github.io/ifs-terminal/update-site/eclipse/rdi8.0/
RDi 9.6 Beta Update Site	https://tools-400.github.io/ifs-terminal/beta-version/update-site/eclipse/rdi8.0/